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Explainable and Scalable Ensemble Decision-Support Framework for Multi-Commodity Price Forecasting in Agricultural Markets

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Abstract

Purpose: The purpose of this paper is to respond to the constant challenge of predicting prices of food commodities in developing economies, which are highly volatile, affecting food security and making policy decisions more difficult. The goal of this study is to develop an end-to-end framework that is reliable, scalable, and complementary to supporting both policy and market interventions in real time.

Methodology: The paper tests a wide range of Machine Learning (ML), Deep Learning (DL), hybrid, and statistical models on a large-scale dataset of 22 commodities across seven states in India. A common benchmarking framework is used to measure performance, and Friedman and Wilcoxon tests are used to determine statistical significance. SHapley Additive exPlanations (SHAP)-based explainability is applied to determine important predictive factors.

Findings: Random Forest (RF) was found to have the best forecasting performance (Root Mean Squared Error (RMSE) = 1.49; $R^2 = 0.9995$), followed by a hybrid Gated Recurrent Unit (GRU)-XGBoost residual model. Classical methods such as SARIMA and Prophet lagged badly. Statistical tests confirmed that tree-based and hybrid models outperform DL and econometric baselines. SHAP analysis revealed that recent lags, rolling means, and volatility measures were the most influential predictors.

Originality/Value: This work provides a comprehensive comparison of forecasting techniques for food commodities in India, demonstrating the practical value of ensemble and hybrid models. The study illuminates their efficacy, transparency, and suitability for real-time deployment for policymakers and supply chain stakeholders.

Keywords: Commodity price forecasting, Agricultural economics, Ensemble learning, Sustainability, Shapley Additive exPlanations analysis, Statistical validation.

1 | Introduction

Food commodity price forecasting is a major issue in agricultural economics, policymaking, and supply chain management. Volatility in the prices of basic commodities has several implications. It tends to have far-reaching consequences for farmers, consumers, and governments, particularly in developing economies,

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where both food security and price inflation control are national agendas. Commodity forecasting has been addressed since earlier times using classic time-series approaches such as ARIMA, SARIMA, and Prophet, which are able to model seasonality and trends [1], [2]. However, these methods have limitations when applied to nonlinear, non-stationary, and highly volatile datasets common in agricultural markets.

Over the past couple of years, the idea of Artificial Intelligence (AI) and Machine Learning (ML) models has become a powerful alternative [3–5]. Ensemble-based predictive models such as Random Forest (RF) and XGBoost are highly predictive, as they capture non-linear relationships in large datasets [3], [6]. In addition, neural networks, such as Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU), are especially effective at learning sequential dependencies in price series, outperforming conventional statistical models in most cases [4], [5]. Further, combination approaches combining statistical decomposition and AI models have been found to yield promising results in addressing the nonlinear and non-stationary duality in commodity price forecasting [7]. Although very common, traditional econometric forecast models such as ARIMA and SARIMA have significant weaknesses when used to predict agricultural commodity prices. These models are based on linear relationships and the assumption of weak stationarity, which are frequently violated in real-life agricultural markets by sudden price increases, nonlinear demand-supply relationships, and policy or weather shocks. Consequently, ARIMA-based models tend to generate overly smooth predictions and take a long time to adapt to abrupt changes in the market. Despite SARIMA being an extension of ARIMA in that it models seasonality, it is still incapable of capturing complex cross-temporal dependencies and heterogeneous regional behavior. Similarly, although Prophet is flexible in dealing with various seasonal trends, it does not respond to short-term volatility as well as other spectral techniques and is less appropriate for high-frequency and shock-sensitive agricultural price sequences.

Despite these developments, three major gaps remain: 1) most existing studies focus on single commodities or spatially isolated markets, which prevents them from generalizing, 2) few studies have provided a rigorous head-to-head evaluation of statistical, ML, Deep Learning (DL), and hybrid methods on the same dataset, and 3) there is a dearth of detailed analysis using multi-state data in India, where the diversity in agricultural production and consumption patterns at the regional level is a decisive factor in determining the price levels of commodities.

This study makes three main contributions to the commodity price forecasting literature: 1) it presents one of the richest empirical assessments to date by being the first to use a large-scale dataset of seven states in Western India and 22 essential food commodities, 2) it compares a wide range of forecasting methods (ML, DL, hybrid, and traditional econometric methods) in a standardized experimental framework, which allows for a holistic assessment of their relative strengths and weaknesses, and 3) it combines strong statistical validation techniques to establish the significance of observed performance differences and simultaneously draws out policy-relevant implications in terms of market stabilization, supply chain management, and consumer protection. Combined, these contributions link methodological rigor with application, placing the study as both an advance of knowledge and a tool for decision-makers. The core contribution lies in integrating large-scale ensemble and hybrid forecasting with rigorous statistical validation and explainability for multi-commodity agricultural price monitoring.

Thus, this combination of various methodologies, multi-state coverage, and strong statistical tests adds not only to furthering the research literature in forecasting but also to the application of forecasting to decision-making in agricultural economics.

2 | Literature Review

Recent studies on food commodity price forecasting include DL (LSTM, GRU, Convolutional Neural Network (CNN), Recurrent Neural Network (RNN)), ML (XGBoost, RF, LightGBM), statistical approaches (ARIMA, SARIMA, Holt-Winters, Prophet), and some hybrid models. A major issue is the nonlinear, non-stationary behavior of price series, which has led to the development of decomposition and hybrid approaches [7–9].

DL methods frequently outperform statistical baselines, as shown for rice in Indonesia [10], spinach in Pakistan [11], vegetables in Beijing [12], and cotton volatility [13]. Enhanced LSTM variants achieve strong results in India [14], [15], U.S. markets [16], and Italy [17], though at a higher computational cost. GRU and its variants have also proven effective, as in Myanmar [18] and Indonesia [19].

The accuracy is enhanced even with hybrid structures, for example, ARIMA -LSTM in Indonesia and Malaysia [20], [21], STL-FA-AttGRU for hog prices [22], and Holt-Winters-GRU for rice [9]. Comparative experiments also show that these hybrids outperform ARIMA, SARIMA, and Prophet on larger training sets [23], [24].

The ML solutions are competitive. XGBoost and RF give good results across India [25], Europe [26], and global datasets [27]. Comparative strategies for ensemble support enhance performance in Indian vegetables [28].

Other innovations include AttLSTM-ARIMA-BP for corn in China [29] and RNNs outperforming SVR for Indian crops [30]. Studies also show practical applications, such as sales and price forecasting in superstores [31] and single-cell LSTM for food and fuel markets [32].

Despite these advances, most studies remain commodity- or region-specific, focusing on rice, vegetables, or livestock in Asia, Europe, and the U.S., limiting generalizability. Only a few large-scale cross-country or multi-commodity studies exist [16], [26], [27].

To fill this gap, the current study conducts a thorough evaluation of state-of-the-art models across 22 commodities and 7 Indian states, including rigorous statistical testing and benchmarking. *Table 1* summarizes the recent literature with an emphasis on methodological diversity, accuracy metrics, preprocessing methods, and computational efficiency. The conclusion from this synthesis is that although hybrid and DL models have frequently outperformed traditional methods in terms of prediction accuracy, their relatively high computational complexity makes scalability an important consideration in practical applications.

Table 1. Overview of recent studies on food commodity price forecasting, highlighting methodological families, preprocessing techniques, adaptability across regions, and computational trade-offs.

Study	Forecast Accuracy Metrics	Data Preprocessing Techniques	Algorithm Adaptability	Computational Efficiency
[9]	Holt-Winters-GRU hybrid achieves RMSE 27.75 with robust generalization	Holt-Winters decomposition for seasonality	Global rice price forecasting	Efficient seasonal decomposition combined with GRU
[10]	LSTM achieves the lowest RMSE (129.67) vs. ML models with high errors	No complex preprocessing; focus on historical price data	Rice retail prices in West Java, Indonesia	LSTM is computationally intensive; SARIMAX is faster but less accurate
[11]	LSTM outperforms ARIMA and RF with R^2 0.8 and the lowest MAE	Data cleaning and preprocessing on large datasets	Spinach prices in Pakistan	Moderate computational cost with big data tools
[12]	LSTM outperforms CNN and ML models in vegetable price forecasting	Standard time series preprocessing	Beijing vegetable markets	Moderate training time
[13]	Multivariate LSTM outperforms GARCH in volatility prediction	Denoising reduces market noise	Cotton commodity volatility	Moderate to high complexity

Table 1. Continued.

Study	Forecast Accuracy Metrics	Data Preprocessing Techniques	Algorithm Adaptability	Computational Efficiency
[14]	Stacked LSTM achieves the best accuracy in cumin price forecasting	Daily price data with multiple accuracy metrics	Cumin prices in Gujarat, India	Moderate to high training time
[15]	LSTM outperforms ARIMA and RNN in groundnut price forecasting	Monthly wholesale price data	Groundnut prices in India	Moderate training time
[16]	DL improves multi-step forecasts relative to the USDA baseline at longer horizons.	Standard time series preprocessing	U.S. agricultural commodities	High computational cost for long horizons
[17]	CNN-LSTM hybrid outperforms ARIMA and Prophet in accuracy, but is slower	Standard time series preprocessing	Italian food wholesale prices	Hybrid requires more tuning time
[18]	GRU with sigmoid activation outperforms LSTM and BiLSTM in RMSE and R^2	Window size and activation function tuning	Daily salt, oil, and rice prices in Myanmar	Efficient training with GRU; LSTM is less consistent
[19]	LSTM slightly outperforms GRU with MSE ~ 0.00035 in food commodity prices	Standard preprocessing with time series normalization	Shallots and chicken meat prices in Banyumas, Indonesia	Moderate training time with standard DL models
[20]	Hybrid ARIMA-LSTM achieves 92.5% accuracy, outperforming ARIMA and regression.	Basic preprocessing; focus on daily price data	Central Java commodity prices	The hybrid model is more complex but yields better accuracy
[21]	Ensemble ARIMA-LSTM outperforms individual ARIMA and LSTM by RMSE	Uses daily retail price data without advanced preprocessing	Malaysian fruit, poultry, and vegetable markets	Ensemble increases computational load but improves results
[22]	STL-FA-AttGRU hybrid achieves the lowest errors in hog price forecasting	STL decomposition and factor analysis	Chinese hog prices	Moderate to high complexity
[23]	Improved LSTM model achieves $>95\%$ accuracy in crop market trend prediction	Time series and multi-variant feature engineering	Corn, oats, soybean markets	Moderate to high training time
[24]	Composite ARIMA-SARIMA model improves agricultural price forecasting	Grid search for parameter optimization	Agricultural product prices	Moderate computational cost
[25]	XGBoost and RF are effective in crop price prediction	Environmental factors like temperature and humidity	Indian crop prices	Moderate computational cost

Table 1. Continued.

Study	Forecast Accuracy Metrics	Data Preprocessing Techniques	Algorithm Adaptability	Computational Efficiency
[26]	XGBoost achieves an average R^2 of 0.85 for food price forecasting	Technical indicators and historical price data	Food prices in 12 European countries	Efficient ML model with moderate preprocessing
[27]	XGBoost and GRU show strong accuracy across 14 commodities	Extensive feature engineering with technical indicators	Diverse commodities, including natural gas and agricultural products	High computational cost due to many models and feature sets
[28]	Stack ensembles and gradient boosting exhibit variable accuracy across commodities and regions.	Monthly and weekly price data with seasonal adjustments	Tomato and potato prices in Indian states	Ensemble models increase computational demand
[29]	AttLSTM-ARIMA-BP hybrid achieves the best accuracy and robustness	Spatial-temporal factor analysis and a priori algorithm	Corn prices in Sichuan Province, China	High complexity due to multiple model integration
[30]	RNN outperforms SVR in Indian crop price prediction	Standard time series preprocessing	Indian crop prices	Moderate training time
[31]	XGBoost predicts sales volume; LSTM predicts wholesale price for vegetables.	Sales volume and price data analyzed	Superstore vegetable sales data	Moderate computational cost
[32]	LSTM single-cell model achieves RMSE 0.14 and R^2 98.2%	Moving averages and volatility features	Bread, meat, milk, oil, petrol prices	Low training time
[33]	LSTM and GRU outperform ARIMA and SARIMA in coffee price prediction	Incorporates fuel prices and weather data as external features	Focus on the Vietnam coffee market with daily data	Moderate training time with added external variables

The RF and XGBoost ML methods have shown impressive predictive ability by capturing nonlinear relationships and interactions among lagged and engineered features. Nevertheless, their performance is highly reliant on comprehensive feature engineering, and their forecasting abilities would weaken in the absence of relevant exogenous drivers. Moreover, ensemble models are robust yet interpretable. Still, they are constrained by a lack of explanation capabilities without the explicit integration of explainability models, such as Shapley Additive exPlanations (SHAP), limiting their direct applicability to policy-driven decision-making scenarios.

The most commonly used DL models for commodity price forecasting include LSTM and GRU networks, which can learn sequential dependencies. However, several studies have reported significant difficulties in implementing these models in practice. To begin with, both LSTM and GRU networks are data-intensive. They can easily overfit when trained on small amounts of per-commodity time series, particularly in markets with disordered prices. Second, these models are computationally intensive and require hyperparameter tuning and long-term training, both of which are costly. Third, trained as universal models for many diverse domains, DL architectures can struggle with generalization, leading to unpredictable or unreliable performance compared to tree-based ensembles.

To overcome the shortcomings of both standalone statistical and DL models, recent studies have devoted increasing attention to hybrid and ensemble forecasting frameworks. Hybrid models include ARIMA-LSTM, XGBoost-LSTM, and CNN-GRU, which integrate linear decomposition or sequential learning with nonlinear residual modeling to perform long-term prediction and capture short-term variability. CNNGRU hybrids, for example, use convolutional networks to extract local features, recurrent networks to model temporal dependencies, and XGBoostLSTM models to use gradient boosting to mitigate systematic errors in DL network predictions. Although these methods are generally more accurate, their additional architectural complexity and computational cost raise questions about practicality in terms of scalability and real-time implementability, especially in large multi-commodity forecasting environments.

2.1 | Research Gap and Contributions of This Study

Although the existing literature has made significant progress, several gaps remain. Most research is based on a single commodity or a small local market, and thus, the findings cannot be generalized. Furthermore, comparative analyses of all models, including statistical, ML, DL, and hybrid models, are seldom performed within a single experimental study that uses identical datasets and validation procedures. Moreover, the fact that hybrid models enhance accuracy does not imply that many studies address the scalability, computational efficiency, and interpretability of the models, which are the main expectations of any operational decision-support system for agricultural policy and market monitoring. Previous studies have either concentrated on commodity-specific issues or on single regional markets [10], [19], [20], [25], thereby limiting the scope of their findings. Despite the promise of hybrid and ensemble models for improving predictive accuracy [21], these paradigms tend to be computationally demanding at scale and have not been systematically tested. DL methods like LSTM and GRU have demonstrated potential compared to classical econometric techniques [11], [12], [18]. Still, they are often tested on a single dataset and rarely use statistical testing methods to assess the reliability of intra-model differences.

This study advances the literature by addressing these gaps in three ways: 1) it is a large-scale, multi-commodity forecast covering seven states of Western India and 22 major food commodities; hence, it is one of the largest forecasting studies in the Indian context, 2) it compares a wide range of predictive models, such as RF, XGBoost, Hybrid GRU+XGBoost, LSTM, GRU, SARIMA, and Prophet, to which it consistently applies the entire set of commodity and state series to provide an overarching view of their comparative performance, and 3) it uses heavy nonparametric statistical analysis, such as Friedman, Nemenyi post hoc, and Wilcoxon signed-rank tests, so that the observed performance differences are not incidental but are statistically substantial.

3 | Methodology

This section introduces the methodological framework used in this study. This section includes the data used, the forecasting models adopted, the experimental setup, the evaluation metrics applied, and the statistical analyses performed to evaluate and compare the performance of various models. In addition, model explainability techniques were incorporated to better understand the contribution of features to predictions. The general methodological scheme of this study, which combines data preparation, experimental setup, forecasting models, evaluation metrics, statistical validation, and explainability, is summarized in *Fig. 1*.

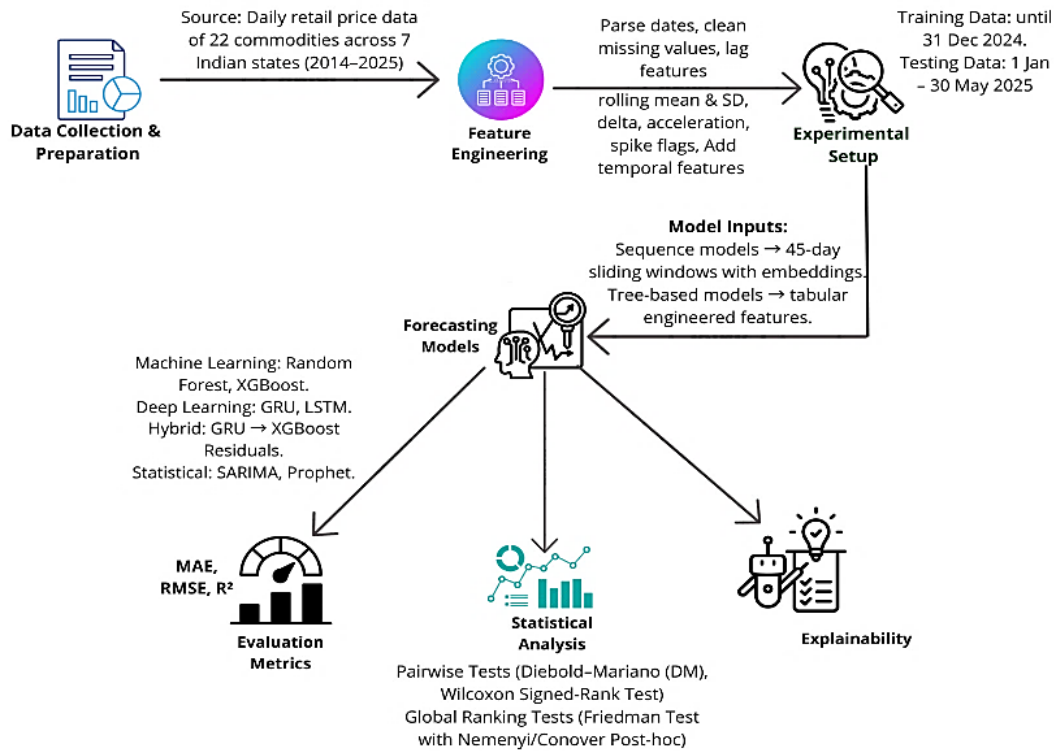


Fig. 1. Methodological framework of the proposed study.

Three main factors influenced the choice of forecasting models in this study: 1) the capability of modeling nonlinear and non-stationary dynamics that are prevalent in agricultural price series, 2) scalability to a large number of heterogeneous commodity-region time series, and 3) appropriateness for practical implementation in data-constrained and resource-limited policy settings. To this end, the modeling framework comprises representative methods from four large methodological families, including classical time-series models (SARIMA, Prophet), tree-based ensemble learning (RF, XGBoost), DL sequence models (LSTM, GRU), and hybrid residual-learning architectures.

3.1| Data Collection and Preparation

The data used in the study is based on the historical daily data of retail price of 22 commodities (Atta (Wheat), Gram Dal, Groundnut Oil (Packed), Gur, Masoor Dal, Milk, Moong Dal, Mustard Oil (Packed), Onion, Palm Oil (Packed), Potato, Rice, Salt Pack (Iodised), Soya Oil (Packed), Sugar, Sunflower Oil (Packed), Tea Loose, Tomato, Tur/Arhar Dal, Urad Dal, Vanaspati (Packed), Wheat) from seven States in Western India, sourced from the publicly available repository¹. The series covers the period from 14 February, 2014 to 11 June, 2025, and provides a long, continuous record of price evolution in the region for several commodities.

Data preparation was conducted systematically in a multi-step manner to make it suitable for both ML and statistical forecasting models. All datasets were initially loaded with CSV files, and the date variable was properly processed to the datetime format to maintain the time sequence of observations. Feature engineering was then used extensively to increase the information content of the series. The lagged price values were generated over 1 to 45 days to capture both short- and medium-term time dependencies. To control for seasonal and calendar effects, calendar-based temporal predictors, such as month, day of the week, and binary weekend indicators, were mined. Moreover, statistical characteristics, including daily price change (delta), acceleration, a 7-day rolling average, and a 7-day rolling standard deviation, were calculated to indicate local trends and price volatility. All lagged, and rolling statistics were built solely on prior observations, with rolling

¹ <https://ceda.ashoka.edu.in/data-portal>

statistics calculated from purely backward-looking windows. The scaling and transformations of the fitted features were applied only to the training data, not to the test data, because they introduce information leakage. In addition to the train-test split, a rolling origin (walk-forward) validation was performed to assess the robustness of the reported results.

Because abrupt market movements were to be explicitly considered, a binary spike flag was added to capture extreme price fluctuations. A spike was defined as an observation in which the change in the price per day was more than two standard deviations of the delta series. Instead of eliminating these observations, this characteristic enables the models to capture shock-related dynamics prevalent in agricultural markets.

The most frequent cause of missing values was the construction of lagged and rolling-window features performed at the start of each time series. Root-wise deletion was used to remove these rows, and the index was restarted to maintain time continuity. The raw price data contained no additional missing values; thus, no imputation methods were used, as imputation can substantially alter the short-term price dynamics of highly volatile series.

Finally, state and commodity identifiers were categorized and added to each observation, and unique numerical codes were generated for use as embedding layers in the neural network models. This end-to-end preprocessing pipeline ensures a clean, application-rich dataset. It preserves the structural features of agricultural price series, enabling a fair and robust comparison of various forecasting procedures.

3.2 | Experimental Setup

The study considers seven forecasting models, namely Hybrid (GRU - XGB residual), LSTM, GRU, RF, XGBoost (XGB), SARIMA, and Prophet, using the daily retail prices of 22 commodities in seven states of Western India. The dataset was split temporally into a training period (up to 31 December, 2024) and a testing period (01 January, 2025 to 30 May, 2025).

XGBoost and RF were selected because they can model nonlinear interactions, handle noisy data, and incorporate engineered temporal features. These ensemble approaches, unlike DL models, work consistently with relatively little per-series data and are less sensitive to hyperparameter tuning. The fact that they performed better in this study indicates that they are appropriate for large, multi-commodity forecasting tasks. The GRU and LSTM models were represented as sequence-learning baselines that could capture temporal dependencies. However, their performance is limited in multi-series environments because they are sensitive to hyperparameters, require more computation, and have limited per-series data. Training a global model with heterogeneous commodities further complicates optimization, resulting in reduced predictive stability.

For the DL models (GRU, LSTM, and hybrid-base GRU), the target variable was optionally log-transformed and standardized using the StandardScaler. The input features were also standardized, and the data were reshaped into input sequences of 45 days. Using embedded layers, state and commodity identifiers were added to account for cross-sectional heterogeneity. Early stopping (patience = 5) was used on a 10% internal validation split to avoid overfitting. Predictions were subsequently inverse-transformed to return the values to the original scale. To demonstrate the selection of the temporal configuration, a sensitivity analysis was performed varying the GRU input window length (30, 45, and 60 days) and the rolling window size used for feature engineering (7, 14, and 21 days). The results indicate that the 45-day input window and 14-day rolling window provide the best compromise between accuracy and stability, and that the choice of configuration is a strong, well-substantiated selection of design parameters for volatile commodity price series.

Tree-based models (RF, XGB, Hybrid residual XGB) were trained on tabular features comprising lagged values, temporal features (month, day of week, weekend), and derived statistics (rolling mean, standard deviation, delta, acceleration, and spike flags). State and commodity identifiers were one-hot-coded. RF was configured with 500 estimators and a minimum leaf size of 5 to achieve reasonable diversity and variance deflation in the ensemble while keeping training time within bounds. Increasing the number of estimators beyond this point did not yield a significant improvement in accuracy at a comparable computational cost.

XGBoost used 1000 estimators, a maximum depth of six, a learning rate of 0.03, a subsample ratio of 0.8, and a colsample by tree of 0.8.

For statistical baselines, time series analysis using historical value series was performed, with SARIMA and Prophet models fitted independently for each state-commodity pair. SARIMA was parameterized with orders (2,1,2) and seasonal components (1,1,1,12), a common practice for modeling monthly data. Prophet decomposed each series into trend, weekly, and yearly seasonality components, with daily seasonality disabled. These implementations focused on the natural time structure of the series, with very few exogenous regressors, as the design assumed was present and ensured comparability. SARIMA-based models were built with fixed autoregressive and moving-average orders and seasonal components in accordance with common empirical practice and are not series-specific, optimized forecasts, but standardized baseline models.

3.3 | Forecasting Models

GRU-XGB is a hybrid (residual) model powered by gradient boosting and RNNs. GRU learns short-term temporal patterns, whereas XGBoost learns the nonlinear relationships between residual errors with engineered features. This design is more accurate than single DL models, and it is also interpretable and scalable. The residual XGBoost model was trained on lagged price features, rolling statistics, volatility, calendar features, and the GRU base predictions to learn systematic residual patterns. DL models were trained on the statistics of the training set, which standardized all numerical inputs. A bidirectional GRU with 64 and 32 hidden units and two layers was used as the base learner. The model was trained using the Adam optimizer (learning rate = 0.001), a custom spike-aware loss (large deviations), dropout (0.3), and L2 regularization (1e4). Residuals from the GRU predictions were then modeled using an XGBoost regressor with engineered features and GRU outputs, and the final forecast was obtained by adding the GRU and residual predictions. The GRU component was optimized using a spike-aware loss that imposes heavier penalties for large prediction errors during extreme price movements. This design is well-suited to high volatility because it does not smooth out price shocks; instead, it focuses on accurately predicting prices.

The LSTM model had the same architecture and optimization scheme, but substituted GRU layers with LSTM layers, serving as a complementary recurrent baseline.

The RF and XGBoost models used the engineered tabular feature space. RF used bootstrap aggregation with 500 trees, whereas XGBoost implemented boosting with optimized hyperparameters of depth, learning rate, and feature subsampling.

SARIMA and Prophet models were used as classical statistical baselines. SARIMA explicitly models autoregressive, differencing, and moving-average components with seasonality, whereas Prophet uses a decomposition-based framework to handle irregularities, outliers, and trends. Both approaches were kept close to their canonical forms to benchmark their inherent strengths against ML/hybrid models.

Table 2 summarizes the methodological rationale behind each modeling component, demonstrating how the baseline, ensemble, DL, and hybrid approaches collectively capture the linear, nonlinear, and sequential price dynamics within a unified framework.

Table 2. Summary of methodology and model selection.

Component	Model/Method	Rationale/Key Contribution
Baseline Evaluation	ARIMA, SARIMA	Serve as classical linear benchmarks capturing trend and seasonality; included for methodological completeness and fair comparison.
	Prophet	Decomposable time-series model handling trend and multiple seasonalities; used as a widely adopted baseline for commodity forecasting.
	RF	Robust ensemble learner effective for nonlinear relationships and tabular data; performs well with engineered lag and rolling features and limited tuning.
	XGBoost (XGB)	Gradient boosting framework with strong regularization; efficiently models nonlinear feature interactions and residual structures.
	LSTM, GRU	DL sequence models capable of learning temporal dependencies; included to evaluate neural approaches under a global multi-series setting.
Hybrid base model	Global GRU	Captures short- and medium-term temporal dynamics across heterogeneous commodity–state series; computationally more efficient than LSTM while retaining comparable sequential modeling capability.
Hybrid residual model	XGBoost	Models residual errors from the GRU base model to correct systematic nonlinear patterns, particularly price spikes and volatility-driven deviations.
Feature utilization	Lag, Rolling, Volatility, Calendar Features	Encodes short-term memory, local trends, seasonality, and market volatility, all of which are essential to agricultural price dynamics.
Performance ranking	Post-hoc Analysis	Confirms that the Hybrid Residual (GRU+XGBoost) model achieves the lowest average rank, statistically outperforming individual baselines across commodities.
Framework contribution	Integrated Ensemble + Hybrid + XAI Pipeline	Combines scalable forecasting, residual correction, and explainability to support decision-making in multi-commodity agricultural markets.

3.4 | Spike-Aware Loss Function

Prices of agricultural commodities often exhibit large, sharp changes that disproportionately affect market monitoring and intervention policies. The standard is similar to the Mean Squared Error (MSE) in that it penalizes all errors uniformly and can thus bias it towards underemphasizing errors that occur during critical spike events. To remedy this, a spike-conscious loss is used to train the GRU part of the hybrid model.

The Spike-aware loss is designed based on the weighted MSE, in which the weight on prediction errors linked to severe price variations is greater:

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^N w_i (y_i - \hat{y}_i)^2,$$

where y_i and $\hat{y}_i$ denote the true and predicted values for sample i , respectively, and w_i is a sample-specific weight. The weight is defined as

$$w_i = \begin{cases} 2.0, & \text{if } |y_i - \mu_y| > \tau, \\ 1.0, & \text{otherwise,} \end{cases}$$

where μ_y is the mean of the target variable, and τ is a predefined deviation threshold identifying potential price spikes.

The threshold τ is also calculated on the same basis as target scaling in model training. In particular, the nominal deviation of the original price scale is normalized using a method similar to that of the target variable

(log transformation and MinMax scaling). It ensures that spike identification is not lost in the neural network's scaled space.

The spike-aware loss is more robust to volatility by imposing a larger penalty for large errors during extreme price movements, and it tends to encourage the model to more accurately predict abrupt market changes. This design aligns with the learning objective and the practical value of price-shock forecasting in agricultural markets.

3.5 | Evaluation Metrics

The performance of the various models was assessed based on several complementary error measures. The Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) were used to capture absolute and squared deviations, respectively, and the RMSE was assigned a higher weight to larger deviations. The proportion of explained variance was calculated using the coefficient of determination (R^2), and the error was measured using the symmetric Mean Absolute Percentage Error (sMAPE), which is scale-independent in time series of small values. In addition, the Mean Absolute Scaled Error (MASE) was used as a metric to compare model forecasts to naive one-step forecasts, with values below 1 indicating that the models outperformed the naive model.

These measures were calculated separately at the state-commodity level and combined to yield general measures of the model's total performance. These numerical comparisons were augmented by formal statistical tests to determine whether the observed differences in performance were statistically significant. The details of this statistical analysis are presented in the following subsections.

3.6 | Statistical Analysis

To obtain rigorous statistical evidence on the relative performance of the forecasting models, several statistical tests were conducted using error metrics (RMSE, MAE, and sMAPE) across commodities. These tests were modeled to determine not only the pairwise differences but also the global performance ranking of all models. First, pairwise tests of the Diebolt-Mariano (DM) with Newey-West variance-adjusted were carried out for every model combination at the commodity level. The null hypothesis tested against evaluates the predictive performance of the two models using squared or absolute forecast errors. To increase interpretability, the results were summarized as the number of commodities in which one model significantly performed better than the other (at the 5% significance level).

Second, Wilcoxon signed-rank tests were used to compare the models' performance across commodities using a nonparametric approach. These tests were used to determine whether the median difference in the accuracy metrics (RMSE, MAE, and sMAPE) between the two models was statistically significant across the dataset. To minimize false-positive inflation due to multiple testing, the resulting p-values were adjusted using the Benjamini-Hochberg False Discovery Rate (BH-FDR). Third, to evaluate differences in global ranking among the models, the Friedman test was used to analyze the ranks of RMSE values across commodities. If the Friedman test revealed a significant difference ($p < 0.05$), post-hoc pairwise comparisons were performed using the Nemenyi and Conover tests, with the Holm correction to control the family-wise error rate. To facilitate visualization, the Critical Difference (CD) was drawn, showing the average ranks of the models, and models that could not be differentiated at the specified significance level were grouped.

The final step involved creating a pairwise win matrix to illustrate the relationships among all competing models, both in terms of model rivalry and overall model accuracy for each pair of models in each commodity. The win matrix shows which model had a strictly lower RMSE in how many commodities. A combination of these analyses, including multiple layers of statistical testing, such as time-series specific tests of model accuracy, nonparametric median model comparisons of model median errors, a global ranking of the models, and graphic summaries, allows for a comprehensive and statistically defensible assessment of the superiority of one model over another in practice.

Nonparametric statistical tests have been assumed to provide a strong basis for comparing homogeneous price series across commodities. The Friedman test was applied to compare the global performance of models using ranks, without assuming normality, which is often violated in time-series forecast errors. The Wilcoxon signed-rank test was used to evaluate the pairwise differences in median error, and it is very robust to outliers. To account for time dependence in forecast errors at the series level, the DM test was used. To reduce multiple testing false positives, p-values were adjusted using the Benjamini-Hochberg False Discovery Rate (FDR) procedure. However, the tests fail to reflect economic relevance and may be influenced by cross-series dependence; hence, the statistical findings are viewed as supportive but not as evidence of the model's superiority.

3.7 | Explainability

To provide additional insights into the interpretability of the results, SHAP analyses were performed on the RF, XGBoost, and residual components of the Hybrid model. SHAP values provide information about the contribution of each feature to the output prediction of each of the three models, individually and collectively; therefore, they provide insights into each model's decision-making process. For a representative subsample of test cases, SHAP analyses produced combined SHAP plots to evaluate the magnitude and direction of the impact of the most important features in each model on predicted prices. Additionally, a ranked table of the top features, based on mean absolute SHAP values, was created to translate the model's complexity into economic terms.

4 | Results and Discussion

This section provides a comparative analysis of the seven forecasting models against various states and commodities. The results are presented as a combination of overall performance metrics, series-level analyses, statistical significance tests, visual interpretations of predictive accuracy, and analyses of the model's interpretability using SHAP values.

4.1 | Overall Performance Metrics

Table 3 reports the aggregated performance of all models across the test dataset using multiple error measures, including MAE, RMSE, R^2 , and sMAPE. RMSE, the primary evaluation metric, ranked the models.

Table 3. Overall, test set performance metrics across all states and commodities.

Model	MAE	RMSE	R^2	sMAPE
RF	0.1911	1.4914	0.9995	0.1348
Hybrid	0.4567	1.8029	0.9993	0.4577
XGB	0.6764	2.1637	0.9990	0.5845
GRU	1.7842	5.1655	0.9942	1.9621
LSTM	1.9041	6.0068	0.9921	1.9237
SARIMA	6.1873	10.4599	0.9762	8.5197
Prophet	13.6282	20.3065	0.9102	15.5123

As shown in *Table 3*, the RF model yielded the most accurate price forecast, with the lowest RMSE (1.4914), smallest MAE (0.1911), highest R^2 (0.9995), and smallest sMAPE (0.1348). Therefore, the RF model demonstrated strong reliability in forecasting prices. The Hybrid GRU-XGB residual model achieved the second-best results, with an RMSE of 1.8029 and an R^2 of 0.9993. Although the Hybrid model was less accurate than the RF model, it still demonstrated competitive results. Finally, the XGBoost model ranked third, with an RMSE of 2.1637 and an R^2 of 0.9990.

In contrast, the DL models (GRU and LSTM) recorded notably higher error values (RMSE=5.1655 and 6.0068, respectively) and lower R^2 values (0.9942 and 0.9921, respectively). Their performance was broadly comparable but markedly weaker than that of the tree-based and hybrid approaches. Traditional time series models performed the worst: SARIMA reached an RMSE of 10.4599 and an R^2 of 0.9762, whereas Prophet was the least effective, with an RMSE of 20.3065 and an R^2 of 0.9102. These findings indicate that statistical

models struggle to capture the complex and heterogeneous dynamics of this multi-series dataset. In contrast, ensemble and tree-based approaches provide substantially more accurate and robust predictions.

The mathematical properties of the evaluation measures can help explain differences in reported performance. RMSE discourages deviations in the forecasting of large proportions and is thus sensitive to volatility and price spikes, whereas MAE encapsulates common deviations of the absolute type. The lower RMSE and MAE values of the Hybrid GRU and XGBoost models obtained after each run indicate better performance under both normal conditions and extreme price changes. The high R^2 values reflect the strong short-run autocorrelation in commodity prices; however, meaningful differentiation among models is often evident in error metrics (RMSE, MAE, sMAPE). Therefore, the high R^2 values indicated by the RF, XGBoost, and the Hybrid models should be interpreted cautiously. These values are also largely motivated by the high autocorrelation in commodity prices and by the successful exploitation of strictly backward-looking lag and rolling properties. To avoid information leakage, all feature transformations were trained only on the training data, and predictions were made based on temporally ordered splits. In addition to the predetermined train-test evaluation, a rolling origin (walk-forward) validation was conducted and yielded similar results, suggesting that the reported results are not due to overfitting. Therefore, error-related measures such as RMSE, MAE, and sMAPE are the primary indicators of forecasting quality.

Tree-based ensemble models are effective at reducing variance through aggregation and nonlinear interactions among lagged and rolling features, resulting in lower generalization error. In contrast, DL models exhibit more dispersed errors, indicating that they are sensitive to hyperparameters and to low effective sample sizes when trained as global models over heterogeneous commodity state series.

4.2 | Ablation Study: Effect of Group Aggregation

To determine the extent to which group aggregation affects performance forecasting, an ablation study was conducted comparing models that explicitly exploit shared learning across commodities and states with those that do not. *Table 4* presents the summarized aggregated metrics (MAE, RMSE, R^2 , and sMAPE) of all state-commodity series.

Group aggregation models, including Global GRU, Global LSTM, and Hybrid Residual (GRU+XGBoost), are trained to learn the common underlying temporal structures of two or more commodities using a shared global modeling architecture and (where possible) with embedding-based conditioning. Among them, the Hybrid Residual model exhibited the best performance (MAE = 0.468, RMSE = 0.682), which indicates that residual learning significantly enhances the robustness of global DL models in heterogeneous multi-series settings. The GRU and LSTM global models also achieved higher results than the classical statistical baselines, which proves the advantage of shared learning over per-series modeling.

Table 4. Ablation results comparing models with and without group aggregation across all states and commodities.

Model	Category	Overall MAE	Overall RMSE	Overall R^2	Overall sMAPE
RF	Without group aggregation	0.258992752	0.568454406	0.999913532	0.296337011
HybridResidual (GRU+XGB)	With group aggregation	0.468114305	0.682252589	0.999875447	0.720862141
XGBoost	Without group aggregation	0.497729988	0.755636646	0.999847212	0.617300223
Global GRU	With group aggregation	1.182183577	1.858811803	0.999075442	1.63237484

Table 4. Continued.

Model	Category	Overall MAE	Overall RMSE	Overall R ²	Overall sMAPE
Global LSTM	With group aggregation	1.387736308	2.118373265	0.998799207	1.829706112
SARIMA	Without group aggregation	6.428726774	11.22752241	0.96626886	8.808749567
Prophet	Without group aggregation	15.96005346	23.95493496	0.846448823	16.01517269

Models that do not explicitly support aggregate the firm behave in a mixed way. The overall error (MAE = 0.259, RMSE = 0.568) for the RF was the lowest and demonstrated the effect of rich lag-based feature engineering on pooled data, but not on latent-group modeling. XGBoost is also competitive (RMSE = 0.756). Still, classical statistical models (SARIMA, RMSE = 11.23) and Prophet (RMSE = 23.95) are not competitive at the portfolio level; thus, their scalability is limited, and they cannot leverage cross-commodity information.

In general, the ablation results indicate that explicit group aggregation is a primary factor that enhances the robustness and scalability of DL-based forecasting models, especially when residual correction is used. Although tree-based models are very competitive with good feature representations, group-conscious models are more advantageous than local statistical models and provide greater stability for global sequence learners in large-stakes multi-commodity prediction.

The explainability component is applied via a post-hoc analysis and does not affect model training or predictive performance. As the contribution of features is sought to be explained, SHAP is implemented on the tree-based models and the residual part of the hybrid architecture, as explained in Section 4.6. Although partial explainability is proven effective within the hybrid framework, full end-to-end explainability is not achievable with the GRU or the residual stage, leaving this for future research.

4.3 | Per-(State, Commodity) Performance

Detailed performance metrics for each individual (state and commodity) series are provided in *Fig. 2*. This granular view reveals the variability in the model performance across different series. This figure shows that the RF model was the best-performing across the individual series, based on RMSE. However, other models, including hybrid models and, in some cases, XGBoost, also demonstrated superiority for specific state-commodity combinations.

4.4 | Visual Analysis of Predictions

Fig. 3 provides a scatter plot of actual versus predicted prices for a subsample of the state's data using all models. The points close to the diagonal line (perfect predictions) are accurate forecasts. The scatter plot confirms the above observation that the predictions of the RF, Hybrid Residual, and XGBoost models are closer to the actual values than those of the DL and traditional time series models, especially for higher price ranges, where the latter tend to deviate more.

Although time series plots have been generated for each of the 22 commodities across the seven Western Indian states, for clarity, *Fig. 4* plots one representative commodity (Atta/Wheat) for each state to show the relative performance of the models. This graph shows the strengths and weaknesses of various models for particular commodities. For example, for some commodities, traditional time series models may produce smoother predictions that miss sharp changes, whereas DL models may exhibit greater volatility in their forecasts.

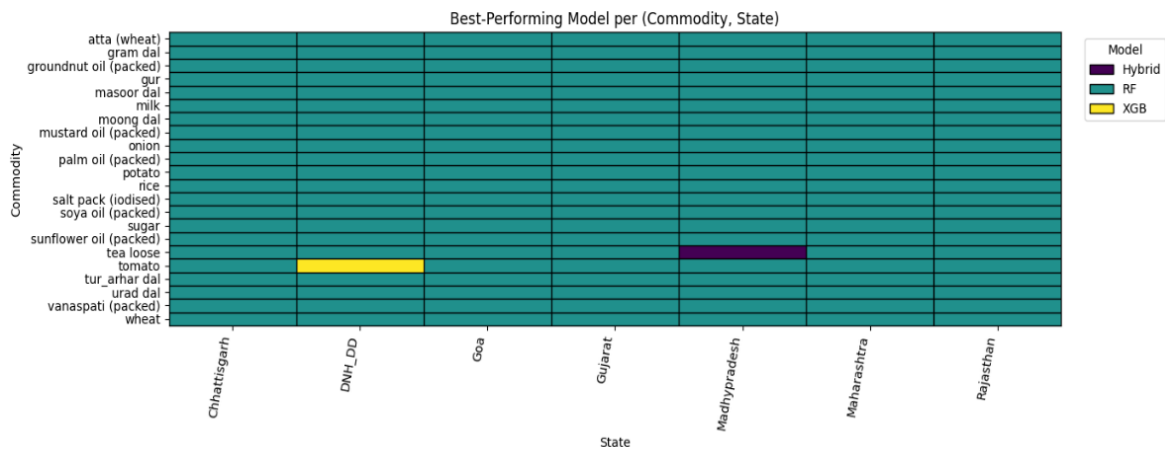


Fig. 2. Heatmap of the best model by (state, commodity) RMSE.

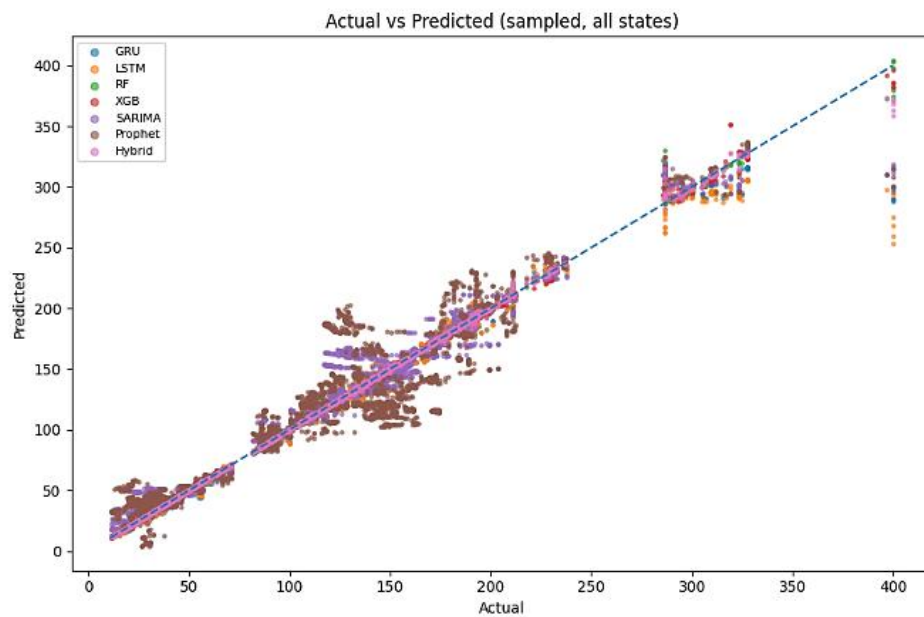
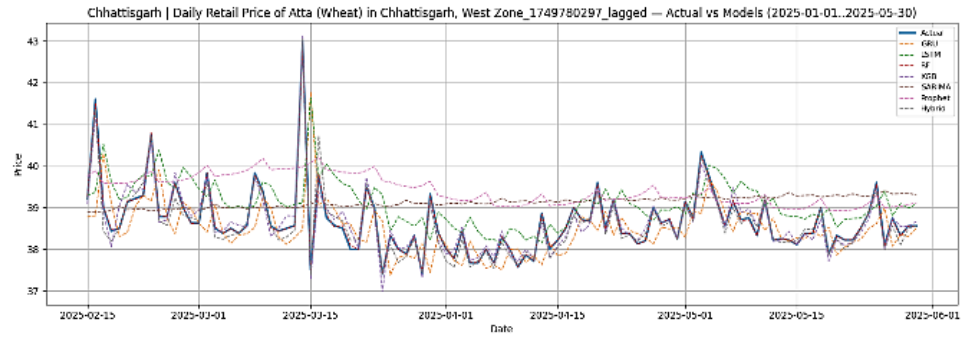
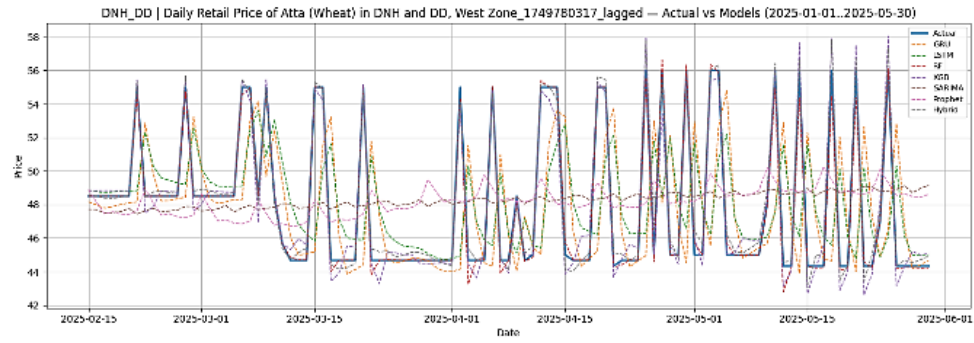


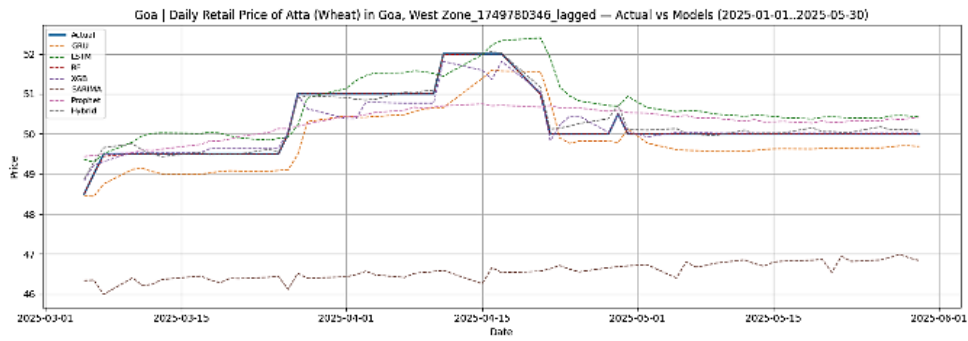
Fig. 3. Scatter plot of actual versus predicted commodity prices across all models (sampled test data, all states).



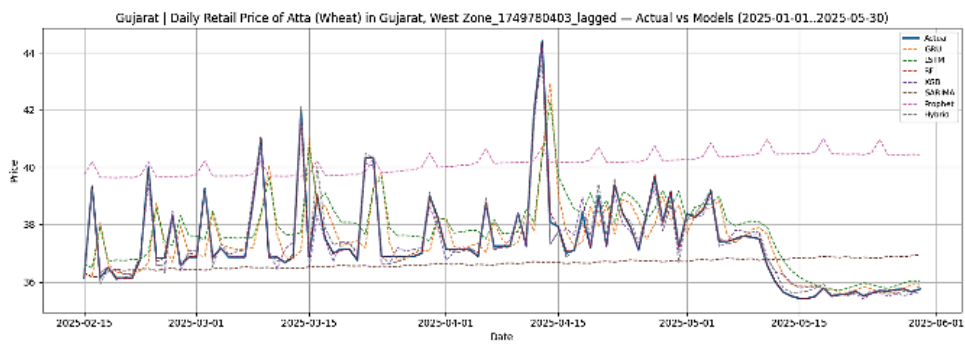
a.



b.



c.



d.

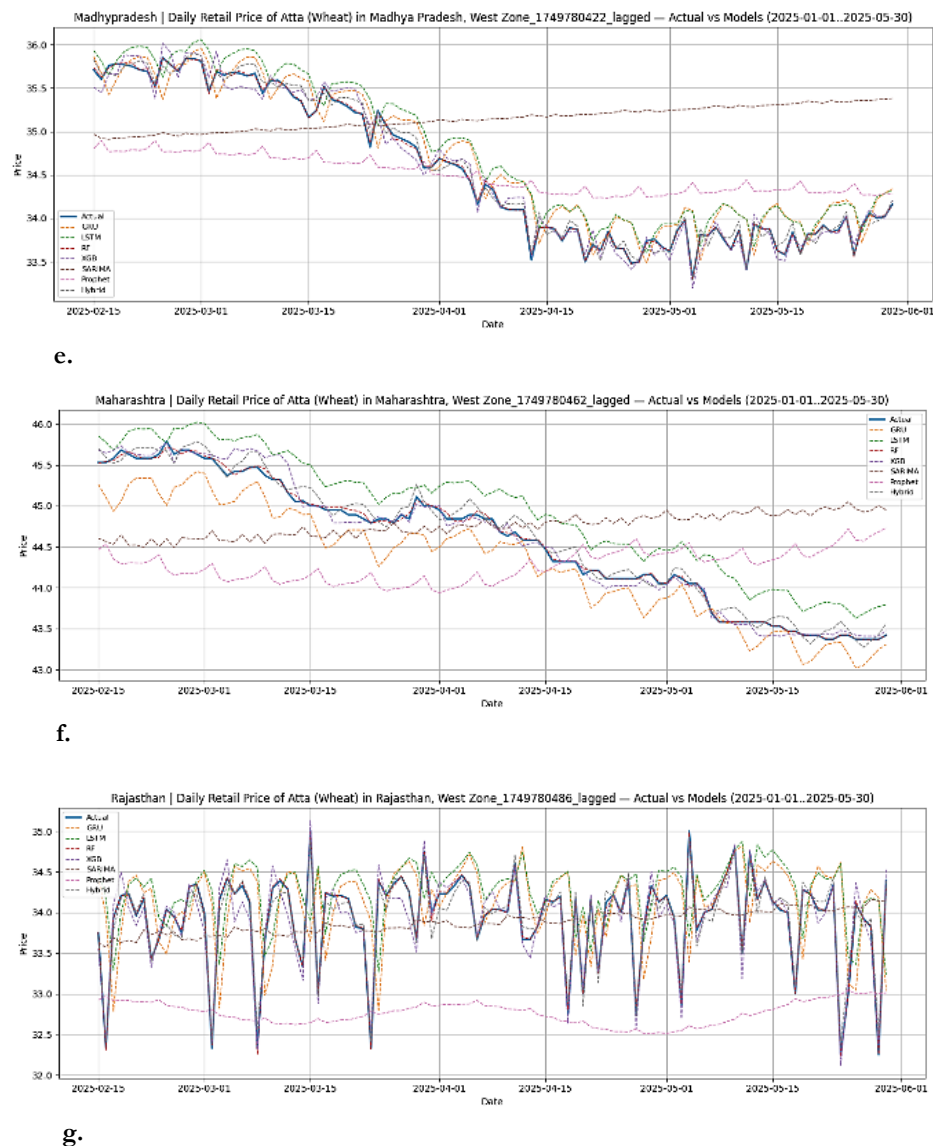


Fig. 4. Time series comparison of actual versus predicted daily retail prices of Atta (Wheat) across different states for the period 2025-01-01 to 2025-05-30, using all forecasting models (GRU, LSTM, RF, XGB, SARIMA, Prophet, Hybrid); a. Chhattisgarh, b. DNH and DD, c. Goa, d. Gujarat, e. Madhya Pradesh, f. Maharashtra, and g. Rajasthan.

4.5 | Statistical Comparison of Models

In addition to the descriptive assessment, a set of statistical tests was applied to the per-commodity model performance measures to formally assess differences between models. Analyses of the relative complementarity between forecast approaches provide strong evidence of a systematic ranking. To rigorously test whether the observed differences in forecasting performance were statistically significant, a series of nonparametric tests was conducted using per-series RMSEs. The Friedman test is used to test the statistical significance of observed performance differences among models when rank distributions are used, with no assumption of normality of forecast errors. Individual series-level tests (Diebold-Mariano) were used to assess the predictive ability of the series, and pairwise differences were assessed using Post-hoc Nemenyi, Conover, and Wilcoxon tests. Combined, these tests can provide collective global and series-level evidence of the solidity of the observed model hierarchy.

The Friedman test, a nonparametric version of repeated-measures ANOVA, was first applied to the ranks of the RMSEs for the state-commodity series. The test yielded a chi-square statistic of [Fr_stat] at a p-value of

[Fr_p]. Since the p-value is less than 0.05, we reject the null hypothesis that there are no real differences in the forecasting accuracy of the evaluated models and conclude that there are statistically significant differences in forecasting accuracy. *Table 5* shows the average rank from the Friedman test, with lower ranks indicating better relative performance.

Table 5. Average ranks of models by RMSE (Friedman Test).

Model	Average Rank
RF	1.0195
Hybrid	2.1883
XGB	2.9935
LSTM	4.5519
GRU	4.8506
SARIMA	5.7403
Prophet	6.6558

In line with the aggregated error metrics, the RF model had the best rank (1.0195), followed by the Hybrid GRU-XGB residual model (2.1883) and XGBoost (2.9935). The DL models (GRU and LSTM) ranked in the middle, and the traditional time series models (SARIMA and Prophet) ranked the lowest. To determine pairwise differences, post-hoc Nemenyi tests with Holm-adjusted p-values were used (*Table 6*).

Table 6. Holm-adjusted p-values from Nemenyi Post-hoc Test on RMSE ranks (values < 0.05 indicate statistically significant differences).

Models	RF	Hybrid	XGB	GRU	LSTM	SARIMA	Prophet
RF	1	4.23E-05	2.24E-14	0	0	0	0
Hybrid	4.23E-05	1	0.018506	0	0	0	0
XGB	2.24E-14	0.018506	1	9.59E-13	5.13E-09	0	0
GRU	0	0	9.59E-13	1	0.889292	0.005599	4.74E-12
LSTM	0	0	5.13E-09	0.889292	1	2.86E-05	3.33E-16
SARIMA	0	0	0	0.005599	2.86E-05	1	0.003771
Prophet	0	0	0	4.74E-12	3.33E-16	0.003771	1

The post-hoc analysis proved that RF and Hybrid significantly outperformed all other models ($p < 0.05$), whereas the difference between RF and Hybrid was not statistically significant. XGBoost was also significantly better than the DL and classical time series baselines, but was still statistically smaller than RF and Hybrid. Notably, there was no significant difference between the GRU and LSTM, consistent with their similar performance. As indicated in *Table 7*, the Conover post hoc test with Holm adjustment provides further evidence that the RF and Hybrid models achieved statistically significant improvements over all other models ($p < 0.001$). The differences between GRU and LSTM were statistically insignificant, indicating similar performance trends for these DL models.

Table 7. Holm-adjusted p-values from the Conover post-hoc test on RMSE ranks.

Models	RF	Hybrid	XGB	GRU	LSTM	SARIMA	Prophet
RF	1	1.20E-39	0.384340077	8.27E-29	3.71E-85	6.00E-14	2.81E-26
Hybrid	1.20E-39	1	2.70E-35	3.57E-112	1.45E-13	4.21E-86	0.009298708
XGB	0.384340077	2.70E-35	1	7.98E-33	1.57E-79	7.60E-17	1.04E-22
GRU	8.27E-29	3.57E-112	7.98E-33	1	2.69E-166	0.000351707	6.71E-93
LSTM	3.71E-85	1.45E-13	1.57E-79	2.69E-166	1	7.39E-139	8.97E-24
SARIMA	6.00E-14	4.21E-86	7.60E-17	0.000351707	7.39E-139	1	4.45E-68
Prophet	2.81E-26	0.009298708	1.04E-22	6.71E-93	8.97E-24	4.45E-68	1

Table 8 presents the results of the Wilcoxon signed-rank test with the Holm correction, which confirm the previous results. RF and hybrid were also significantly better than the competing models (highly significant), and the comparison of GRU and LSTM also showed no difference ($p = 0.021$). These findings also indicate the superior robustness of the RF and hybrid models over the evaluated series.

Table 8. Pairwise Wilcoxon signed-rank test with Holm correction on RMSE values.

Model_A	Model_B	Stat	P_Raw	P_Holm
RF	Hybrid	100	3.53E-26	3.89E-25
RF	XGB	1	5.17E-27	1.06E-25
RF	GRU	0	5.07E-27	1.06E-25
RF	LSTM	0	5.07E-27	1.06E-25
RF	SARIMA	0	5.07E-27	1.06E-25
RF	Prophet	0	5.07E-27	1.06E-25
Hybrid	XGB	1437	3.02E-16	9.07E-16
Hybrid	GRU	3	5.37E-27	1.06E-25
Hybrid	LSTM	0	5.07E-27	1.06E-25
Hybrid	SARIMA	1	5.17E-27	1.06E-25
Hybrid	Prophet	0	5.07E-27	1.06E-25
XGB	GRU	525	9.47E-23	5.68E-22
XGB	LSTM	399	9.69E-24	6.78E-23
XGB	SARIMA	192	2.05E-25	2.05E-24
XGB	Prophet	0	5.07E-27	1.06E-25
GRU	LSTM	4693	0.021504627	0.021504627
GRU	SARIMA	1180	5.83E-18	2.33E-17
GRU	Prophet	316	2.10E-24	1.89E-23
LSTM	SARIMA	1129	2.59E-18	1.30E-17
LSTM	Prophet	339	3.21E-24	2.57E-23
SARIMA	Prophet	1581	2.52E-15	5.04E-15

Beyond rank-based tests, the DM test was used to compare predictive accuracy at the series level. *Tables 9* and *10* summarize the results for the number of state-commodity series for which one model was significantly better than any other model based on absolute and squared forecast errors, respectively (Holm-adjusted $p < 0.05$). As shown in *Tables 9* and *10*, RF significantly outperformed all other models in a substantial number of series in both absolute and squared errors. The Hybrid model also demonstrated a large number of significant wins against GRU, LSTM, SARIMA, and Prophet. The comparisons between RF, Hybrid, and XGBoost show fewer significant differences, with RF and Hybrid winning some series against XGBoost. The DM test provides additional evidence of the better performance of the RF and Hybrid models at the individual-series level.

Table 9. Pairwise DM Test Summary (absolute errors) - Values indicate the number of series where the column model significantly outperformed the row model.

Model_A	Model_B	B_Better_Count_Sig	A_Better_Count_Sig	N_Comm
GRU	LSTM	43	24	154
GRU	Prophet	5	129	154
GRU	SARIMA	10	103	154
Hybrid	GRU	0	145	154
Hybrid	LSTM	0	141	154
Hybrid	Prophet	0	152	154
Hybrid	SARIMA	0	152	154
Hybrid	XGB	3	41	154
LSTM	Prophet	3	133	154
LSTM	SARIMA	9	102	154
RF	GRU	0	153	154
RF	Hybrid	0	145	154
RF	LSTM	0	154	154
RF	Prophet	0	154	154
RF	SARIMA	0	154	154
RF	XGB	0	152	154
SARIMA	Prophet	20	102	154
XGB	GRU	2	134	154
XGB	LSTM	2	122	154
XGB	Prophet	0	150	154
XGB	SARIMA	3	144	154

Table 10. Pairwise DM Test Summary (squared errors) - Values indicate the number of series where the column model significantly outperformed the row model.

Model_A	Model_B	B_Better_Count_Sig	A_Better_Count_Sig	N_Comm
GRU	LSTM	41	16	154
GRU	Prophet	5	122	154
GRU	SARIMA	11	93	154
Hybrid	GRU	0	108	154
Hybrid	LSTM	0	100	154
Hybrid	Prophet	0	147	154
Hybrid	SARIMA	0	142	154
Hybrid	XGB	3	34	154
LSTM	Prophet	3	126	154
LSTM	SARIMA	8	92	154
RF	GRU	0	119	154
RF	Hybrid	0	86	154
RF	LSTM	0	111	154
RF	Prophet	0	148	154
RF	SARIMA	0	145	154
RF	XGB	0	119	154
SARIMA	Prophet	20	99	154
XGB	GRU	2	99	154
XGB	LSTM	2	88	154
XGB	Prophet	0	145	154
XGB	SARIMA	3	136	154

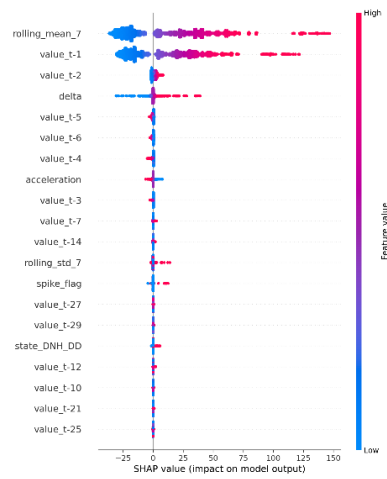
Finally, a pairwise wins matrix (*Table 11*) was created to visually show the frequency with which each model achieved the lowest RMSE among its peers. RF had the highest number of wins among all models, providing further empirical evidence of the model's superior performance.

Table 11. Pairwise wins matrix based on RMSE comparisons across all commodities and states.

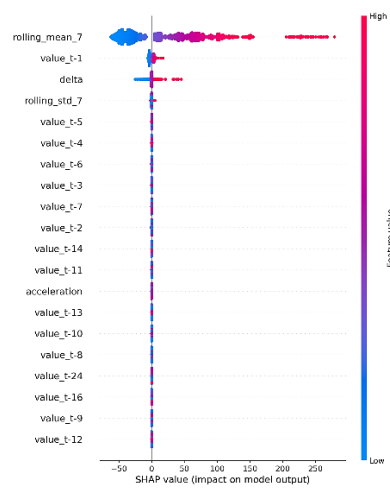
Models	RF	Hybrid	XGB	GRU	LSTM	SARIMA	Prophet
RF	0	152	153	154	154	154	154
Hybrid	2	0	125	153	154	153	154
XGB	1	29	0	143	141	149	154
GRU	0	1	11	0	59	121	139
LSTM	0	0	13	95	0	124	145
SARIMA	0	1	5	33	30	0	125
Prophet	0	0	0	15	9	29	0

4.6 | Shapley Additive Explanations Explainability

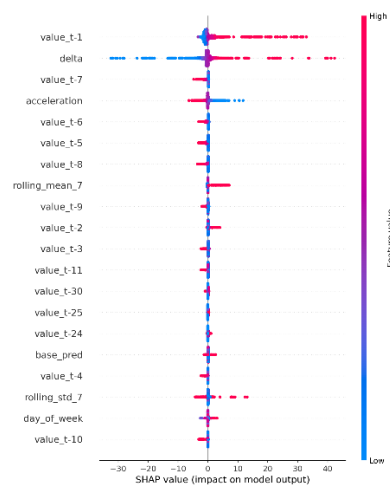
SHAP analysis was performed to gain insights into the feature importance of the tree-based models (RF, XGB, and Hybrid Residual XGB). *Fig. 5* shows the SHAP summary plots for the RF, XGBoost, and the Hybrid Residual XGBoost models.



a.



b.



c.

Fig. 5. SHAP Summary Plot; a. RF, b. XGBoost, and c. Hybrid Residual XGBoost.

The SHAP analysis reveals that for both the direct RF and XGBoost models, the `rolling_mean_7` feature and recent lag features (e.g., `value_t-1` and `value_t-2`) are consistently the most important predictors of commodity prices. It highlights the strong influence of recent price history and local trends on future price changes. Other lag features at different time steps also contributed to the predictions. Delta and acceleration features that capture recent price changes are also important.

For the Hybrid Residual XGBoost model, the SHAP analysis focused on the features that influence the prediction of residual errors from the base GRU model. While recent lag features (value_t-1) and change features (delta, acceleration) remain important for predicting the residual, other features further back in the time series (value_t-7, value_t-6, value_t-8, etc.) and the base_pred itself also appear in the top features, suggesting that the residual model learns to correct for patterns and errors that the base GRU model might have missed, potentially related to more complex or longer-term dependencies.

The one-hot encoded state and commodity features also show some importance in the direct tree-based models, indicating that there are baseline price-level differences or specific patterns associated with different states and commodities that these models capture.

Well-organized economic mechanisms in agricultural markets can explain SHAP outcomes. Recent lag features dominate, indicating significant short-term price inertia stemming from slow supply and inventory adjustments and information-related friction in agricultural value chains. The rolling mean captures local price variations and seasonal persistence driven by planting periods, time to harvest, and regular demand. The characteristics of volatility, including price fluctuations and acceleration, are evidence of supply shocks, weather disturbances, or abrupt demand changes, which are typical in agricultural markets. Combined, these results imply that short-term momentum and local volatility are among the primary drivers of price formation, providing policymakers with the actionable signals needed to intervene early in the market and stabilize it.

4.7 | Discussion

The results of this study provide a comprehensive comparison of several forecasting models for multi-state, multi-commodity daily retail price prediction. Among the evaluated models, the tree-based ensemble methods (RF and XGBoost) and the proposed hybrid model (GRU→XGB Residual) demonstrated superior performance in terms of RMSE, MAE, and sMAPE, outperforming both the DL models (GRU and LSTM) and traditional time series approaches (SARIMA and Prophet).

The superior performance of RF and XGBoost in capturing nonlinear, complex interactions and relationships among engineered features, such as lagged values, rolling statistics, and categorical identifiers, is a major contributor to their success. In addition, the ensemble nature of these models provides robustness, especially in multi-series data, where some series are more influential than others. Furthermore, the Hybrid model, which uses a GRU for sequence learning and XGBoost for residual correction, performs similarly to the RF model, indicating that residual learning compensates for errors in deep sequence models by leveraging tabular features to correct them. Although the RF and Hybrid GRU2XGBoost models showed the most impressive predictive performance, the results should also be evaluated for computational efficiency. The tree-based ensemble model offers a desirable precision-efficiency trade-off: it requires less time to train, is less hyperparameter-sensitive, and scales better to heterogeneous series of commodities and states.

In contrast, the comparatively poorer results of GRU and LSTM may be explained by their greater sensitivity to architectural decisions, the more time-consuming training, and the smaller effective sample sizes per series when trained in the global models. Although the hybrid model is more accurate because it removes GRU residual errors, it is more expensive to use because it uses two stages of training. In terms of deployment, RF can be easily adopted for real-time forecasting and policy monitoring, and hybrid and DL models can be recommended only when higher accuracy margins require more computational power.

On the other hand, GRU and LSTM failed to outperform the tree-based models, demonstrating that in various multi-series contexts, lag, momentum, and volatility engineered features may have greater predictive potential than pure sequence modeling, especially when there is limited per-series data. Additionally, training a single DL model over 154 series ($22 \text{ commodities} \times 7 \text{ states}$) creates optimization challenges and makes the model significantly more sensitive to hyperparameters, contributing to its poor aggregate performance.

Traditional models such as SARIMA and Prophet performed poorly, mainly because they relied solely on autoregressive and seasonal components, without accounting for external features. It highlights the importance of feature-rich approaches for commodity price forecasting, where short-term demand-supply dynamics and regional effects often drive shocks and volatility.

The analysis of the computational costs (Table 12) also aligns with the rankings of observed accuracy, with the RF being the most economical at the expense of predictive power, followed by the hybrid model, which is accurate at the expense of the high cost of training, and the DL models, which reach high predictive power at a large expense in this multi-series environment.

Table 12. Computational cost and model complexity comparison.

Model	Training Time (s)	Inference Time (s)	Trainable Parameters
Global GRU	1350.55	1.37	58,769
Global LSTM	1240.85	1.18	77,649
RF	88.44	0.16	N/A (Tree-based)
XGBoost	4.67	0.01	N/A (Tree-based)
Hybrid Residual (GRU + XGBoost)	1353.72	1.39	58,769 + Trees
ARIMA (Global Est.)	3.98	0.07	~132
SARIMA (Global Est.)	411.26	0.09	~154
Prophet (Global Est.)	22.51	1.02	N/A (Bayesian)

Statistical validation through Friedman, Nemenyi, Conover, Wilcoxon, and DM tests reinforces the observed hierarchy. The RF and Hybrid models offer statistically significant improvements over DL and time-series methods. In contrast, the RF and Hybrid models remain statistically comparable across many pairwise tests. It suggests that both models are strong candidates, depending on the deployment requirements.

SHAP analysis further clarifies the drivers of predictive performance. Features such as short-term lags, rolling means, and volatility indicators (delta, acceleration) were consistently influential, confirming that recent momentum is the dominant signal in commodity price movements. The prominence of state and commodity identifiers demonstrates that fixed regional and commodity-specific effects are key determinants of price levels. For the Hybrid model, the SHAP values of the residual learner confirmed its corrective role, integrating both engineered features and GRU outputs.

4.8 | Practical Implications

These findings suggest that tree-based ensemble and hybrid models are particularly well-suited for deployment in state-level price-monitoring and early-warning systems. Policymakers can leverage their robustness for short-term forecasting to trigger interventions, such as price stabilization, stock release, or supply chain adjustments. The ability of RF and Hybrid to capture heterogeneity across states and commodities makes them well-positioned for real-time monitoring platforms, where reliability and interpretability are critical. The relative underperformance of GRU and LSTM highlights that in real-world multi-series applications, computationally demanding sequence models may not always provide added value when feature engineering can effectively summarize temporal dynamics. It implies that for resource-constrained agencies or developing economies, deploying RF or Hybrid models can achieve high accuracy without requiring a specialized DL infrastructure.

4.9 | Limitations and Future Work

Although the proposed framework performs well for multi-state, multi-commodity retail price forecasting, it has several limitations. The dataset is mostly composed of historical price data. It does not account for other exogenous variables, such as weather conditions, trade policies, and macroeconomic factors, which could further improve predictive power. In addition, all series share a common DL configuration, which may limit the model's ability to capture commodity-specific dynamics fully. Future research can thus consider adding exogenous variables, series-specific or transfer-learning neural networks, and probabilistic forecasting methods to more efficiently measure uncertainty. The empirical analysis is conducted across seven states of Western India. Still, the proposed framework is methodologically universal and could be used in other places that share the same market structure, data availability, and price volatility characteristics [34]. Nonetheless, the transferability of the results may be influenced by differences in institutional settings, market integration, and policy environments. Thus, extending the analysis to larger geographic regions is a significant research direction for the future. Moreover, integrating Transformer-based models and smart optimization, as suggested in the latest literature, can be viewed as another perspective that should enhance long-range dependency modeling [35]. Embedding stochastic simulation blocks into hybrid forecast pipelines also enables uncertainty representation and scenario-driven decision analysis in future developments of this research [36].

5 | Conclusion

In a systematic evaluation that considered ensemble, hybrid, DL, and statistical forecasting in a comparative environment, this study presents an overall comparison of forecasting models of daily retail prices of 22 major food commodities across seven states in India. The results for the ensemble models are very stable and show them to be the best for volatile and multi-series price data. The RF model showed the best overall accuracy, with the lowest RMSE of 1.49, MAE of 0.19, and R^2 of 0.9995, indicating almost perfect predictive ability. The Hybrid GRU-XGB residual model followed closely behind, with an RMSE of 1.80 and an R^2 of 0.9993, indicating the effectiveness of combining sequential learning with residual correction. XGBoost also performed well (RMSE = 2.16, R^2 = 0.9990). In contrast, the DL-based models GRU (RMSE = 5.16, R^2 = 0.9942) and LSTM (RMSE = 6.00, R^2 = 0.9921) achieved less accuracy than the aforementioned models, and the conventional time series models SARIMA (RMSE = 10.45) and Prophet (RMSE = 20.30) achieved the worst results. Using the statistical tests like the Friedman test, Wilcoxon test, Diebold and Mariano test, it was noted that RF and Hybrid models are very powerful over the series. SHAP, a Shapley Additive models to Interpret Predictions (SHAP) explainability method, indicated that short lagged predictors, rolling averages, and volatility were the most important predictors, with state and commodity IDs also being important. Taken together, these results identify ensemble and hybrid models as practical, easily scalable, interpretable, and policy-relevant commodity price surveillance and early warning systems.

Although the empirical results of the suggested framework are strong, this study has several limitations. Its analysis is mainly based on past prices and does not directly incorporate exogenous factors such as weather, policy interventions, or macroeconomic factors that may affect agricultural price dynamics. Moreover, DL models were applied in a unified configuration across commodities, which could impair their capacity to learn series-specific features. Future studies can overcome these limitations by incorporating exogenous variables, using multi-stage and adaptive hybrid architectures, and evaluating models with transformer-based architectures that have a greater ability to capture long-range dependence and interact across commodities. Also, recent developments in data-driven demand prediction, hybrid ML, and optimization reported in the literature on supply chain, agriculture, and commodity analytics [37], [38] further demonstrate the increased credibility of intelligent decision-support systems. These studies support the applicability of predictive modeling, optimization, and explainable analytics to complex, uncertainty-driven industrial settings, which supplement the rationale and results of the current study.

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Author Contribution

Conceptualization, R. J.; methodology, R. J., N. L., R. D. and A. O.; software, R. J., N. L.; validation, R. J., N. L., R. D. and A. O.; formal analysis, R. J.; writing-creating the initial design, R. J.; writing-reviewing and editing, R. J.; visualization, R. J., N. L., R. D. and A. O.. All authors have read and agreed to the published version of the manuscript.

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Data Availability

The dataset used in this study is publicly available on Kaggle and can be accessed from the “CEDA Daily Food Prices Data. <https://dca.ceda.ashoka.edu.in/index.php/home/timeseries>”

Conflicts of Interest

The authors declare no conflict of interest.

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چارچوب پشتیبان تصمیم‌گیری تجمیعی تبیین‌پذیر و مقیاس‌پذیر برای پیش‌بینی قیمت چندین کالا در بازارهای کشاورزی

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چکیده

هدف: هدف این مقاله، پاسخ به چالش مداوم پیش‌بینی قیمت کالاهای غذایی در اقتصادهای در حال توسعه است که به شدت بی‌ثبات هستند و بر امنیت غذایی تأثیر می‌گذارند و تصمیم‌گیری در مورد سیاست‌ها را دشوارتر می‌کنند. هدف این مطالعه، توسعه یک چارچوب جامع است که قابل اعتماد، مقیاس‌پذیر و مکمل برای پشتیبانی از مداخلات سیاستی و بازار در زمان واقعی باشد.

روش‌شناسی پژوهش: این مقاله طیف گسترده‌ای از مدل‌های یادگیری ماشین، یادگیری عمیق، ترکیبی و آماری را بر روی یک مجموعه داده در مقیاس بزرگ از ۲۲ کالا در هفت ایالت هند آزمایش می‌کند. یک چارچوب معیار مشترک برای اندازه‌گیری عملکرد و آزمون‌های فریدمن و ویلکاکسون برای تعیین اهمیت آماری استفاده می‌شود. قابلیت توضیح مبتنی بر SHAP برای تعیین عوامل پیش‌بینی‌کننده مهم اعمال می‌شود.

یافته‌ها: مشخص شد که جنگل تصادفی بهترین عملکرد پیش‌بینی را دارد ($RMSE=1.49$ و $R^2=0.9995$) و پس از آن یک مدل باقیمانده ترکیبی GRU-XGBoost قرار دارد. روش‌های کلاسیک مانند SARIMA و Prophet به شدت عقب مانده‌اند. آزمون‌های آماری تایید کردند که مدل‌های مبتنی بر درخت و ترکیبی بهتر از یادگیری عمیق و خطوط پایه اقتصادسنجی هستند. تحلیل SHAP نشان داد که تاخیرهای اخیر، میانگین‌های غلتان و معیارهای نوسان، تأثیرگذارترین پیش‌بینی‌کننده‌ها هستند.

اصالت/ارزش افزوده علمی: این کار به ارائه مقایسه‌ای جامع از تکنیک‌های پیش‌بینی کالاهای غذایی در هند کمک می‌کند که ارزش عملی مدل‌های ترکیبی و گروهی را نشان می‌دهد. این مطالعه، اثربخشی، شفافیت و مناسب بودن آن‌ها را برای استقرار در زمان واقعی جهت استفاده توسط سیاست‌گذاران و ذینفعان زنجیره‌تامین روشن می‌کند.

کلیدواژه‌ها: پیش‌بینی قیمت کالا، اقتصاد کشاورزی، یادگیری جمعی، پایداری، تحلیل شکل، اعتبارسنجی آماری.